Peripheral — a memory of what you've seen on screen

Hermes Agent Plugin Hackathon entry · dossier for verification · 5 Nov 2026 · `okyssta/hermes-peripheral` v1.0.1

1. The entry in one paragraph

Hermes can read a file you hand it and can look at the screen *now*. It cannot answer for a screen that is already gone — *“that error I got yesterday”*, *“the price on that page”*, *“what was on screen when it broke”*. Peripheral indexes the screenshots the OS already writes, reads them locally with Tesseract, and gives the agent four tools: `screen_recall`, `screen_timeline`, `screen_show`, `screen_index` — plus a `/peripheral` command and a bundled skill. The timeline puts the user's screens and the agent's own tool calls on one clock. Local only: no network calls, no capture daemon, secrets redacted at ingest — in the OCR text and in the agent-event log alike. `/peripheral forget` deletes the row and the thumbnail this plugin wrote for it; your screenshots are never touched.

2. Claim ledger

STATUS	CLAIM	EVIDENCE
`demonstrated`	The pipeline indexes real screenshots and finds the screen that answers a natural question.	10 real 1440×900 pages (5 captured live from Stripe/GitHub/AWS/HN/Python docs, 5 rendered local apps) → 10.3 s, 12,616 chars stored, 1.3 MB of source images; query `"card error"` returns screen #8 and the line *“The card has expired. Check the expiration date or use a different card.”* A warm rescan re-reads zero images.
`demonstrated`	The model reaches for the tools unprompted and quotes them correctly.	Captured session (`docs/demo-transcript.txt/.json`): `deepseek-v4.1-flash`, given the four tool schemas and only *“what was that card error I saw on Monday morning?”*, chose 5 calls in 27.4 s — `screen_recall(card error)`, `screen_recall(card declined)`, `screen_show(8)`, `screen_timeline(2026-11-02)`, `screen_show(9)` — and answered with `expired card`, its quoted sentence, and the note that the other Monday screens were a Slack thread and a pytest run. The video renders this scene from the transcript.
`demonstrated`	Admissible by the catalog's own gate.	`hermes plugins validate` at the pinned code commit `25342ff`: manifest, config schema, capability probe, declared tools/hooks match, security scan **safe**, no core override. **27/27** tests, including forget-removes-thumbnail, event-detail redaction and the `before` bound.

`demonstrated`	No catalog entry, no repo in the competition wave, and no community-directory entry does screen history.	978 catalog entries parsed; nearest are `mediarag`, `document-reader`, `source-tray`, `modlens`, `hermes-time-machine` — each differing in mechanism (§4). 480-repo wave (since 1 Sep) grepped for screen/screenshot/recall/timeline.
`negative`	The in-gateway chat demo could not be recorded on this host.	`hermes chat -q` fails with `invalid_params` for *new* sessions regardless of this plugin — reproduced with the plugin disabled, on a slash command, and on a second profile. The four tool schemas were verified separately against the provider (HTTP 200, correct tool selection), and the plugin loads/registers under the real loader.
`hypothesis`	"Useful for ~90% of users."	Every screenshot-taker has this question; screenshots are a default OS behaviour (macOS Desktop, Windows Snipping Tool, Linux folders) and the plugin also indexes images the user sends the agent. Users who never screenshot get nothing until they do — that is the honest limit of the claim.
`planned`	Submission steps 2–3.	X post tagging `@NousResearch` and the Typeform — both need your account; copy is ready on request. Catalog PR is open (step 1 done).

3. What was built

Surfaces

- 4 tools (toolset `peripheral`), 1 slash command, 1 observer hook (`post_tool_call`), 1 skill
- Settings in `plugin.yaml` render in Desktop → Settings → Plugins
- State in `<HERMES_HOME>/plugin-data/peripheral/` via the documented storage API

Mechanics

- Incremental scan by mtime+size+SHA-256; near-duplicates dropped at Hamming ≤ 3 on a 64-bit dHash
- Tesseract as one bounded subprocess (25 s cap, one thread); no OCR ⇒ visual index instead of failure
- SQLite FTS5 (porter) + BM25, re-ranked by recency and text richness; query sanitised into quoted prefix terms
- Evidence snippet re-chosen from the stored text — the line with the most query terms, windowed around them

Measured on this machine (2 vCPU ARM, no GPU)

```
10 screens (1440×900)  →  10.3 s   (1.03 s/screen)
text stored            →  12,616 chars
warm rescan            →  0 images re-read
tests                  →  27 passed
validate               →  security scan: safe
captured session       →  5 tool calls, 27.4 s
video                  →  74 s, 0.8 MB, silent
```

Privacy posture

- Zero network calls; no telemetry, no sync, no model call to index
- Never captures the screen — reads files the OS already wrote
- `/peripheral forget <id>|all` deletes the row **and** the thumbnail this plugin wrote; `pause`; `retention_days`

- Redaction: `sk-`/`ghp_`/`AKIA`/`AIza`/Slack, JWT, bearer/basic, `password:`, credentials in connection URLs, private keys, card numbers, 32+ hex — applied to OCR text and to stored agent-event details
- The screenshots themselves are never modified or deleted — only copies the plugin made

4. Prior-art audit (the part that decides “unique”)

, `hermes-vision-inline`

NEAREST THING	WHAT IT ACTUALLY DOES	WHY THIS IS NOT THAT
`mediarag`	Hybrid semantic search over a document/media corpus you point it at (285 MB embeddings, MCP).	No screen capture, no time semantics, no timeline; a library indexer.
`document-reader`	OCR of documents into a Desktop viewer, per profile.	Reads what you give it; no history, no agent tools, no search by time.
`modlens`	Vision bridges that describe an image to a text-only model.	Per-image perception; nothing persists, nothing is searchable.
`source-tray`	Records the URLs the *agent* opened, in a Desktop pane.	Captures the agent's browsing, not the user's screen, and stores no content.
`hermes-time-machine`, `session-diff`	Timeline/scrubber over *sessions*.	Conversation history, not the screen.
Windows Recall, screenpipe (22k★)	OS-level screenshot search / open computer history.	Not agent-native: no tools, no time-bounded recall, no join with agent actions, not installable in Hermes. They prove the demand.

The delta is one mechanism: a persistent, time-indexed percept of the user's screen that the agent can query, joined to a log of what the agent itself did. Scope of the claim: “no catalog entry and no repo in the wave does this”, not “no software anywhere does this”.

5. Deliverables

— also sent to Discord

ARTIFACT	WHERE
Plugin source (MIT)	`https://github.com/okyssta/hermes-peripheral`
Catalog PR	`https://github.com/NousResearch/hermes-agent/pull/149463`
Demo video (74 s, silent)	`docs/demo.mp4`
Captured session	`docs/demo-transcript.txt` · `.json` — the Q&A the video renders from
Tests / harness	`tests/` (27), `scripts/demo_session.py`, `scripts/make_demo_video.py`
Installed locally	`main` profile, watch dir `/root/demo-screens/Screenshots` (10 screens indexed)

The catalog pins the code commit `25342ff` (source, tests, demo, transcript). This PDF is committed immediately after it: `git diff 25342ff HEAD -- '*.py'` is empty, so what you read describes exactly what is pinned.

6. Review round 1 — what changed, and the test that holds it

An internal review read the code cold and reproduced three defects. Each is fixed in `25342ff`, with a test that fails on the old behaviour. Nothing else in the submission changed its claim status.

DEFECT	FIX
`/peripheral forget` deleted rows but left the 360 px copy of the screen on disk.	`db.forget` unlinks the thumbnail it wrote and sweeps the thumbs tree on `forget all`. Reproduced live: 10 thumbs / 41,692 bytes → 0, user's originals untouched. Tests: `test_forget_deletes_the_thumbnail_it_wrote`, `test_forget_all_sweeps_every_thumbnail`.
Agent event details were stored verbatim, so `--password=` and bearer tokens reached the events table.	Redaction moved to the storage boundary (`db.record_event`) and applied in the hook before truncation; a pattern for credentials inside connection URLs was added. Tests: `test_agent_event_details_are_redacted_before_storage`, `test_post_tool_call_hook_redacts_before_the_event_is_written`.
The “model chose the tools” claim cited a script that sent the literal `Bearer ***`, and the video's answer text was hardcoded.	The script uses the key, writes `docs/demo-transcript.json` / `.txt`, and the video renders its Q&A scene from that transcript; it refuses to render without one. The Q&A in §2 is the captured run. Evidence: the transcript itself.

Minor items also fixed: `before <time>` was searched as a lower bound (now an upper bound, with a test), and the settings docstring pointed at a test file that did not exist.

7. What I would do next, in order

- Post the X demo and file the Typeform (needs your account) — the only two submission steps left.
- Add a capture adapter per OS behind an explicit opt-in, if the catalog review asks for one; deliberately withheld for v1.
- Semantic fallback for paraphrased queries (FTS only today): embed the OCR text when a local model is available, keeping text search as the deterministic base.
- Re-record the demo through `hermes chat` once this host's new-session provider error is fixed.

Prepared by Hermes (kanban task t_a960d4f7) for Kyssta · every number above came from a command run on this host · claims labelled demonstrated / hypothesis / negative / planned, no unlabelled assurances.